

ALL-RUSSIAN SCIENTIFIC SCHOOL «YOUNG MEDICINE»

NOMINATION:

Biomedical Engineering and Digital Health Technologies

PROJECT TITLE:

**IN SILICO ORGANOID TWINS: AUTOMATED BIOINFORMATICS SIMULATION
FOR PATIENT-SPECIFIC DRUG SCREENING**

Project Type: Engineering/ Business domain

Organization:

[Biversity School of Biosciences]

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TABLE OF CONTENTS

TABLE OF CONTENTS	2
PROJECT ABSTRACT	3
1. RELEVANCE OF THE TOPIC	4
2. PROJECT OBJECTIVE	6
3. PROJECT TASKS	7
Task 1: Data Infrastructure and Preprocessing Pipeline	7
Task 2: Multi-Scale Organoid Simulation Engine	7
Task 3: Model Calibration and Validation	7
Task 4: Web Application Development and Clinical Integration	7
4. IMPLEMENTATION TIMELINE	9
5. STAGES OF THE PROJECT	10
Stage 1 — Data Infrastructure (Month 1)	10
Stage 2 — Organoid Geometry Modeling (Month 2, Part 1)	10
Stage 3 — Cellular Dynamics Simulation (Month 2, Part 2)	10
Stage 4 — Drug Modeling and Diffusion (Month 3)	10
Stage 5 — Calibration and Validation (Months 4–6)	11
Stage 6 — Web Application and Deployment (Months 7–9)	11
6. THE PROJECT PRODUCT	12
6.1 Core Simulation Engine	12
6.2 Streamlit Web Application	12
6.3 Beneficiaries and Value Delivered	12
7. THE CUSTOMER OF THE PROJECT	14
7.1 Pharmaceutical and Biotechnology Companies	14
7.2 Contract Research Organizations (CROs)	14
7.3 Government Health Agencies and Research Foundations	14
8. CONCLUSION	15
LIST OF SOURCES USED	16

PROJECT ABSTRACT

Patient-derived organoids (PDOs) represent a breakthrough in personalized oncology, achieving 80–90% correlation with clinical drug responses compared to 50% for conventional 2D cell cultures. However, their widespread adoption in high-throughput pharmaceutical screening is severely constrained by scalability limitations: manual culturing yields only 10^3 – 10^4 organoids per batch over several weeks, with batch variability of 20–50% and per-test costs exceeding USD 10,000. These barriers translate into billions of dollars lost annually due to late-stage drug attrition and delayed therapeutic development.

This project proposes the development of an automated bioinformatics pipeline — In Silico Organoid Twins — that computationally replicates patient-specific organoid behavior using multi-scale modeling. The system integrates: (i) U-Net-based segmentation of patient biopsy images for organoid topology reconstruction; (ii) transcriptomic data and literature-informed parameters integrated into agent-based models to simulate heterogeneous cell population dynamics; (iii) quantitative structure-activity relationship (QSAR) and pharmacokinetic/pharmacodynamic (PKPD) modeling via RDKit and OpenEye for drug diffusion and metabolism; (iv) calibration against publicly available pharmacogenomic datasets (e.g., DepMap, GDSC) and organoid studies reported in the literature; and (v) GPU-accelerated computation using PyTorch enabling high-throughput virtual drug screening using GPU-accelerated computation.

The prototype will be delivered as a Streamlit web application with FHIR integration, outputting dose-response curves (IC50 values, drug synergy maps) from inputs of next-generation sequencing (NGS) data and histological images. Projected outcomes include a 95% reduction in per-test cost (USD 100 vs. USD 10,000), turnaround reduction from weeks to hours, and aiming to achieve high concordance with experimental PDO response trends. The system will enable personalized oncology screening, reduce drug development attrition by 30%, and support sovereign pharmaceutical development in resource-limited or sanction-affected settings through local high-performance computing (HPC) deployment.

Keywords: patient-derived organoids, in silico modeling, drug screening, multi-scale simulation, single-cell RNA sequencing, QSAR, pharmacokinetics, personalized oncology.

1. RELEVANCE OF THE TOPIC

The global oncology drug development pipeline faces a systemic crisis: over 90% of cancer drug candidates fail in clinical trials, resulting in estimated losses of USD 1–5 million per compound per year in false-positive screening costs alone [1]. A primary cause is the inadequacy of preclinical models. Traditional two-dimensional (2D) cell cultures fail to replicate the complex architecture, heterogeneity, and microenvironmental interactions of *in vivo* tumors, achieving only ~50% sensitivity in predicting clinical drug responses [2].

Patient-derived organoids (PDOs) — self-organizing three-dimensional (3D) structures cultivated from patient tumor or stem cells — represent a paradigm shift. PDOs achieve 80–90% sensitivity in drug response prediction, better capture tumor microenvironment (TME) effects, and have demonstrated 30% superior KRAS inhibitor response prediction in colorectal cancer over conventional models [3]. Despite these advantages, organoid technology remains inaccessible to most pharmaceutical and contract research organizations (CROs) due to critical scalability bottlenecks.

Manual organoid culturing yields only 10^3 – 10^4 organoids per batch, requires 4–6 weeks per experimental cycle, and costs upward of USD 10,000 per test panel. Batch-to-batch variability of 20–50% further undermines reproducibility, eroding trust in multi-center studies. For pharmaceutical companies evaluating hundreds of compounds per patient cohort, these constraints make PDO-based screening operationally and economically infeasible.

The practical value of solving this scalability problem is enormous. A computational equivalent to physical organoid testing would: (1) eliminate the physical throughput ceiling, enabling simultaneous virtual testing of unlimited compound libraries; (2) reduce per-test costs by 95%; (3) compress experimental timelines from weeks to hours; and (4) democratize access to high-fidelity drug screening for institutions lacking specialized laboratory infrastructure. In regions such as Russia, where post-sanctions constraints limit access to cutting-edge biomedical equipment, an HPC-deployable *in silico* system would directly accelerate sovereign pharmaceutical development.

The convergence of advances in deep learning, multi-scale biological modeling, and GPU-accelerated computing has made this project technologically feasible today. Tools such as U-Net architectures for biological image segmentation [4], CellBlender for agent-based cellular simulation, and physics-informed neural networks for partial differential equation (PDE) solving provide a

mature technical foundation. The present project integrates these capabilities into a unified, validated, and clinically deployable pipeline.

2. PROJECT OBJECTIVE

To develop, validate, and deploy an automated multi-scale bioinformatics pipeline — In Silico Organoid Twins — that replicates patient-specific organoid drug responses computationally, aiming to approximate organoid-level drug response behavior while significantly reducing computational cost and turnaround time compared to experimental workflows, at 1% of the cost and in 1% of the time, enabling high-throughput personalized oncology drug screening.

3. PROJECT TASKS

The following four primary tasks define the project implementation logic and correspond directly to the stages described in Section 5:

Task 1: Data Infrastructure and Preprocessing Pipeline

Establish a robust data ingestion and preprocessing system capable of handling next-generation sequencing (NGS) outputs (scRNA-seq, whole-exome sequencing) and histological biopsy images. This includes FHIR-compliant data parsing, quality control filtering, cell type annotation, and image preprocessing for downstream modeling. Deliverable: a reproducible Python pipeline (BioPython + Scanpy) processing raw clinical inputs into simulation-ready formats.

Task 2: Multi-Scale Organoid Simulation Engine

Build the core computational engine comprising: (i) U-Net-based organoid geometry reconstruction from histological images; (ii) scRNA-seq-driven agent-based models (PhysiCell / CellBlender) simulating large-scale cell population dynamics with biologically informed parameters including proliferation, apoptosis, and intercellular signaling; (iii) QSAR/PKPD models (RDKit, OpenEye) for drug molecule property prediction; and (iv) PDE solvers for drug diffusion, metabolism, and spatial concentration gradients within the organoid volume.

Task 3: Model Calibration and Validation

Calibrate and validate all simulation components against public pharmacogenomic datasets (e.g., DepMap, GDSC) and curated organoid studies. Validation metrics will include: Pearson correlation of predicted vs. experimental IC50 values, area under the dose-response curve (AUC), and drug synergy prediction accuracy. Target: $\geq 95\%$ correlation with experimental PDO outcomes. Deliverable: validated model ensemble with documented performance benchmarks.

Task 4: Web Application Development and Clinical Integration

Develop and deploy a user-facing Streamlit web application enabling clinicians and researchers to submit patient genomic and histological data and receive dose-response curve outputs (IC50, synergy scores) within hours. Integrate FHIR API support for Electronic Health Record (EHR)

system compatibility. Implement GPU-accelerated inference (PyTorch) for high-throughput screening of 10^5 virtual compounds per hour. Deliver documentation, API reference, and a pilot deployment on local HPC infrastructure.

4. IMPLEMENTATION TIMELINE

The project is structured across a 9-month implementation horizon divided into three major phases: development (3 months), validation (3 months), and deployment with clinical piloting (3 months). The table below provides a detailed schedule:

Phase	Period	Key Activities	Deliverable
Month 1	Weeks 1–4	Data pipeline; FHIR parser; scRNA-seq preprocessing	Preprocessing module
Month 2	Weeks 5–8	U-Net segmentation; agent-based model construction	Geometry + Cell modules
Month 3	Weeks 9–12	QSAR/PKPD models; PDE solver integration; GPU optimization	Full simulation engine
Month 4–5	Weeks 13–20	Calibration on DepMap/OrganoidDB; statistical validation; benchmark reporting	Validated model + report
Month 6–9	Weeks 21–36	Streamlit web app; FHIR integration; HPC deployment; pilot testing with clinical partner	Deployed web application

The development phase (Months 1–3) focuses on constructing and integrating all simulation modules. The validation phase (Months 4–6) calibrates models against experimental PDO data and benchmarks performance. The deployment phase (Months 7–9) delivers the clinical-facing web application and completes pilot integration with a collaborating oncology center. Total project duration: 9 months from project initiation.

5. STAGES OF THE PROJECT

Stage 1 — Data Infrastructure (Month 1)

Establish the data ingestion and preprocessing pipeline. Raw inputs include NGS files (FASTQ/VCF formats) and histological whole-slide images (WSI). The pipeline performs: quality control (FastQC, MultiQC), read alignment (STAR aligner for scRNA-seq), cell type annotation (Seurat/Scanpy), and image normalization for U-Net training. A FHIR R4-compliant API module will parse clinical metadata. All code will be version-controlled (GitHub) and containerized (Docker) for reproducibility.

Stage 2 — Organoid Geometry Modeling (Month 2, Part 1)

Train and deploy a U-Net convolutional neural network on annotated histological biopsy images to segment organoid topology — extracting spatial cell distributions, lumen architecture, and stromal boundaries. Training data: publicly available annotated organoid image datasets (>5,000 images). Output: 3D mesh representations of patient-specific organoid geometry for downstream cellular simulation.

Stage 3 — Cellular Dynamics Simulation (Month 2, Part 2)

Integrate single-cell RNA sequencing data into an agent-based model (ABM) using the PhysiCell framework. Each simulated cell agent is parameterized by its transcriptomic profile (gene expression, pathway activity, cell cycle state). The ABM simulates large-scale cell populations, modeling: proliferation/apoptosis dynamics, cell-cell communication (ligand-receptor interactions), and mechanical forces within the organoid geometry obtained in Stage 2.

Stage 4 — Drug Modeling and Diffusion (Month 3)

Model drug behavior within the virtual organoid using a two-layer approach. First, QSAR models (RDKit, OpenEye) predict molecular properties: lipophilicity, membrane permeability, metabolic stability, and protein binding affinity. Second, PDE solvers (FEniCS framework) simulate drug diffusion, convection, and metabolic consumption across the organoid's spatial geometry, generating time-resolved intracellular concentration profiles for each agent cell.

Stage 5 — Calibration and Validation (Months 4–6)

Calibrate the full pipeline against 1,000+ PDO drug response profiles from DepMap and OrganoidDB. A Bayesian optimization framework (Optuna) will tune model hyperparameters to minimize deviation from experimental IC50 measurements. Validation will use 5-fold cross-validation and an independent holdout dataset. Performance targets: Pearson $r \geq 0.95$, AUC-ROC ≥ 0.92 for drug response classification. Results will be compiled into a peer-review-ready benchmarking report.

Stage 6 — Web Application and Deployment (Months 7–9)

Develop the clinical user interface using Streamlit. The application accepts clinician-submitted NGS and histology data, queues simulation jobs on GPU-accelerated HPC nodes (PyTorch inference), and returns interactive dose-response curves (IC50, Bliss synergy scores) within 1–4 hours. The system is containerized via Docker/Kubernetes for scalable deployment. A pilot integration will be conducted with one oncology center to validate clinical usability and data security compliance.

6. THE PROJECT PRODUCT

The primary project product is a validated, deployed In Silico Organoid Twin Platform — an integrated computational system comprising three interconnected components:

6.1 Core Simulation Engine

A Python-based multi-scale modeling pipeline (BioPython, PhysiCell, RDKit, FEniCS, PyTorch) that processes patient genomic and histological data and generates virtual organoid drug response predictions. The engine is modular, allowing independent updates to each simulation layer. It operates on GPU-accelerated hardware and supports parallel execution of 10^5 virtual drug screens per hour.

6.2 Streamlit Web Application

A web-based clinical interface accessible via standard browser, requiring no specialized software installation. Features include: secure data upload (NGS files, histology images), real-time simulation progress tracking, interactive dose-response curve visualization, downloadable reports in PDF and CSV formats, and FHIR-compliant EHR integration for seamless clinical workflow embedding. The interface is designed for use by oncologists and clinical pharmacologists without bioinformatics expertise.

6.3 Beneficiaries and Value Delivered

- Pharmaceutical companies and CROs — receive 95% cost reduction in organoid-equivalent drug screening (\$100 vs. \$10,000 per test) with throughput scaled from hundreds to millions of virtual tests, accelerating compound prioritization and reducing attrition.
- Oncologists and clinical researchers — receive within-hours personalized drug sensitivity profiles for individual patients, enabling evidence-based treatment selection and clinical trial stratification.
- Public health systems and sovereign healthcare programs — gain access to an HPC-deployable, locally operable screening platform independent of imported biological reagents or specialized equipment, supporting drug import-substitution programs.
- Academic research institutions — benefit from an open validation dataset and benchmarking framework, advancing the field of computational pharmacology.

7. THE CUSTOMER OF THE PROJECT

The primary potential customers and product owners are:

7.1 Pharmaceutical and Biotechnology Companies

Large pharmaceutical companies and mid-size biotechs conducting oncology drug development programs represent the highest-value commercial customer segment. These organizations spend USD 1–5 million annually on false-positive screening costs and face 6–12 month delays from inadequate preclinical models. The In Silico Organoid Twin platform directly replaces or supplements their early-stage compound screening infrastructure. Companies such as Biocad (Russia), R-Pharm, and international firms with Russia-market operations represent a target customer base for licensed deployment.

7.2 Contract Research Organizations (CROs)

CROs providing drug screening services to pharma clients face competitive pressure to reduce turnaround times and improve predictive accuracy. Adoption of the platform as a service offering would allow CROs to reduce setup costs (currently USD 50,000+ for organoid panel establishment), eliminate 4–6 week experimental timelines, and differentiate their service offering with in silico capabilities.

7.3 Government Health Agencies and Research Foundations

National health agencies responsible for sovereign pharmaceutical development (e.g., the Russian Ministry of Health, the Russian Science Foundation) represent institutional customers with a mandate to accelerate domestic drug development. The platform's capacity for local HPC deployment without dependence on imported biological materials aligns with post-sanctions import-substitution priorities.

8. CONCLUSION

This project addresses a clearly defined and economically significant bottleneck in oncology drug development: the inability to scale patient-derived organoid drug testing to meet the throughput demands of modern pharmaceutical screening. By constructing a validated, multi-scale *in silico* representation of patient-specific organoids — integrating genomic data, 3D cellular dynamics, and drug physicochemistry — the In Silico Organoid Twin platform provides a computationally equivalent yet dramatically more scalable and cost-effective alternative to physical PDO experimentation.

The practical significance of the project results is substantial and quantifiable:

- A 95% reduction in per-test cost enables pharmaceutical-grade personalized drug screening in settings previously limited to elite research institutions.
- Compression of experimental timelines from 4–6 weeks to 1–4 hours dramatically accelerates preclinical development cycles.
- 95% predictive accuracy relative to experimental PDO data, validated on 1,000+ public datasets, provides a scientifically rigorous foundation for clinical translation.
- A 30% projected reduction in drug attrition translates to billions of dollars saved across the pharmaceutical industry and faster delivery of effective therapies to patients.
- Local HPC deployment capability supports sovereign pharmaceutical development agendas, reducing dependence on imported biological reagents and international laboratory services.

The materials developed during this project — the validated simulation engine, the annotated training dataset, and the benchmarking framework — constitute a foundational resource for the emerging field of computational organoid pharmacology. Prospects for further development include: extension to additional cancer types and non-oncology indications, integration of immunological microenvironment simulation for immuno-oncology applications, regulatory pathway development for clinical decision-support approval, and commercialization through SaaS licensing or CRO partnership models.

The project represents a technically feasible, scientifically rigorous, and commercially viable solution to one of the most pressing scalability challenges in modern drug development.

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